

Pulsed Interferometry Towards Real Time PON Monitoring with Feeder and Drop Fibres Discrimination

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Abstract *We present a non-invasive interferometric method for simultaneous and unambiguous monitoring of PON drop fibres. Drop fibre discrimination is demonstrated with a 30 km feeder fibre deployed in Athens, even with 1:32 splitting ratio and 52 dB budget losses. ©2026 The Author(s)*

Introduction

Passive optical network (PON) systems are susceptible to various types of failures: in particular, service disruptions caused by fibre cuts or infrastructure damages (e.g. cabinets) can lead to significant financial losses for operators and service providers [1]. Legacy PONs OpEx and infrastructure integrity can be improved by in-service real time monitoring of the optical distribution network (ODN). Currently optical time-domain reflectometry (OTDR) is employed for in-service monitoring in PONs; however, it localizes failures only after the impairment has already affected services. Moreover, it is largely limited to detecting feeder fibre breakpoints. More generally, the point-to-multipoint (P2MP) architecture of PONs introduces significant challenges, arising from both high signal losses and the one-to-many topology, which results in crosstalk among the backscattered signals from different drop fibres. Nowadays, the sensing technology that has met greatest success is distributed acoustic sensing (DAS) in the form of coherent time domain optical reflectometry. It has proven to be excellent in terms of reach, sensitivity, and spatial resolution in point-to-point configurations [2], but its use in P2MP architectures is not straightforward. In order to circumvent the key challenges posed by PONs, different ideas have been recently presented. In [3], sensing over a PON is demonstrated exploiting the unique signature of “coded” enhanced scatter fibres. Despite its high performance, this approach requires to replace the standard fibres used in today’s networks, hence it is more compatible to future deployments. To achieve orthogonality in DAS operation, wavelength and code multiplexing has been proposed which for sure increases the overall cost of fibre monitoring [4]. Recently, a DAS approach employing optical frequency domain reflectometry (C-OFDR) has been proposed [5] which is able to distinguish all the drop fibres of a PON, and local-

ize the perturbation with a spatial resolution of 1 m. This approach is very effective, however it has the disadvantage of the extremely high cost of the interrogation system, as it requires the use of I/Q modulators, high speed DACs for frequency sweeping, very high sampling rates (Gsa/sec) and demanding processing involving GPUs and high-end embedded systems. Although the aforementioned DAS techniques have demonstrated promising approaches for monitoring PONs, their cost remains a significant barrier to widespread deployment in access networks. Therefore, there is still a clear need for lower-cost solutions that require minimal impact in the existing network infrastructure. In this direction, sensing approaches based on polarimetry [6] have been proposed having the advantage of being cost-effective and minimally invasive. However, they require significant modifications at ONU side.

In this work, we propose a pulse-based interferometric approach that enables independent monitoring of each drop fibre in a P2MP network, such as a PON, by treating the network as an array of Michelson interferometers with slightly different lengths. The pulse width is adjusted to resolve drop fibres with length differences on the order of a few meters. With respect to solutions based on back-scattering, the proposed technique enhances its tolerance to the high losses typically encountered in PONs. The proposed solution leverages wavelength division multiplexing (WDM) to separate the sensing and the data channels, as conventionally applied to current OTDR based monitoring. The optical interrogator is located at the Central Office, while selective mirrors are required at the optical network units (ONUs) to reflect only the sensing wavelength. The approach is minimally invasive, allowing integration into legacy PON infrastructures with only limited modifications. We demonstrate the capability of the proposed approach to simultaneously and unambiguously monitor the feeder fibre and all the drop fibres by identifying unique perturbations

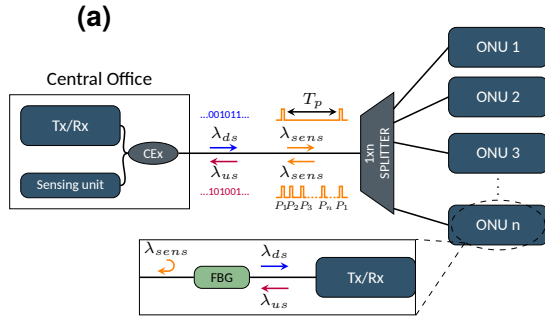
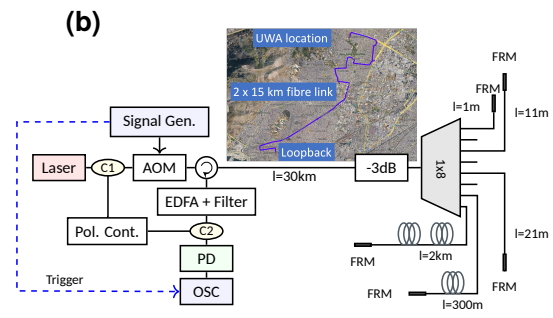


Fig. 1: (a) General scheme of the interferometric sensor integrated onto a PON network. Tx/Rx = Transmitter/Receiver, CE_x = coexistence element, λ_{sens} = sensing wavelength, λ_{ds} and λ_{us} = downstream and upstream wavelengths, respectively; (b) Experimental setup.

per fibre. We showcase the compatibility of the proposed technique to PON ODN higher than 26 dB single pass losses losses (> 52 dB equivalent losses) in the sensing wavelength, in case of a deployed feeder fibre affected by real environmental noise.

Principle of Operation

The concept is illustrated in Fig.1(a). The sensing unit is located at the Central Office and transmits pulses at a wavelength different from downstream and upstream spectral bands. The sensing wavelength is reflected by a wavelength-selective mirror (e.g. a fibre Bragg Grating, FBG) at the ONU side. This is the only modification at the ONU side which has no effect in power consumption or power budget for the communication operation. To ensure minimal overlap between different reflected pulses, the drop fibres must have a nonzero differential length, as is usually the case in real deployments. On one hand, the minimum length difference ΔL_{min} determines the pulse width $T_w \leq \frac{2\Delta L_{min}}{v_g}$. On the other hand, the maximum length difference ΔL_{max} among the drop fibres determines the repetition period $T_p \geq \frac{2\Delta L_{max}}{v_g}$. At the receiver, the inter-arrival delay between pulses enables identification of their corresponding optical paths and allows reconstruction of the associated time-domain phase signal, in a manner similar to per-span interferometry used in subsea cables [7]. Each return pulse integrates the effects accumulated along its propagation path, which includes both the feeder fibre and the corresponding drop fibre. From an interferometric perspective, this defines a distinct interferometric channel, temporally separated from the other $n-1$ channels, as illustrated in Fig.1(a), where $P_1, P_2, \dots, P_n$ denote the reflected pulses from the shortest to the longest path, respectively. Each return pulse integrates the effects of its path consisting of the feeder fibre and its corresponding drop fibre. To isolate feeder fibre effects from drop fibre effects, various methods can be employed, all based on retrieving the feeder fibre's phase contribution and subtracting it from the signals associated with the drop fibres. The first and most effective method involves the use of a ref-



erence interferometry channel loop that accounts only for the perturbations acting on the feeder fibre. A reference channel can be established by connecting an 1×2 coupler to one of the splitter outputs, allowing a small portion of the power to be reflected by the FBG while the remainder propagates to the ONU. The second method consists in the estimation of the contribution of the feeder fibre by performing ensemble averaging of the signals produced by all the available interferometric channels. For simplicity, in this work we directly terminated one output of the splitter to a reflector to obtain the reference channel. The working principle can be extended to more complex architectures such as multi-stage topologies.

Experimental Results

The experimental setup is depicted in Fig.1(b). We use a Koheras E15 laser emitting 16 dBm, at 1550 nm. A 90/10 coupler (C1) splits the light and sends the 90% to an acousto-optic modulator (AOM) which is driven by $T_w = 25 \text{ ns}$ pulses at 20 kHz repetition rate ($T_p = 50 \mu\text{sec}$), while the 10% is used as the local oscillator (LO). With T_w , T_p selected, one can consider $\Delta L_{\min} = 2.5 \text{ m}$ and $\Delta L_{\max} = 5 \text{ km}$. The generated pulsed light is sent into the network through an optical circulator (OC). The P2MP network is composed by a 30 km fibre loop deployed in the city of Athens, acting as the feeder fibre, a 1x8 splitter and 5 drop fibres having a length of 1 m, 11 m, 21 m, 300 m and 2 km, respectively. A 3 dB attenuator is placed before the splitter, to emulate the losses of a 1:16 splitter. The minimum total loss budget per single pass is around 23 dB. Each drop fibre is terminated to a Faraday rotator mirror (FRM) which plays the role of the reflector in our demonstration. For field deployments, FBGs can be used to reflect back just the sensing wavelength. The back reflected pulses travel back the PON reaching the OC; through the OC third port, after pre-amplification (EDFA + Filter)), they are mixed with the LO by means of a 50/50 coupler (C2). A polarization controller (PC) is used to adjust the LO polarization in order to maximize the interference. However, a polarization diversity receiver can be used to acquire all the

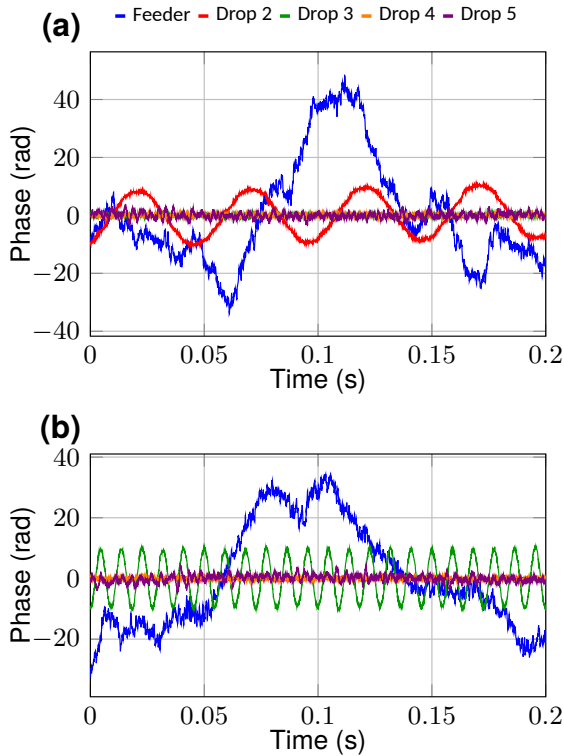


Fig. 2: Examples of the reconstructed phase signals in presence of stretcher induced stimuli in drop fibre 2 (a) and drop fibre 3 (b), and noise affecting the feeder fibre. The perturbation is correctly reconstructed, allowing for accurate identification of the perturbed drop fibre.

power per polarization axes. A single photodiode (PD), is used to receive the signal which is acquired by an oscilloscope (OSC) with 400 MS/s sampling rate. Off-line processing performs the downconversion of the raw signal and the I/Q demodulation after the reconstruction of the low frequency phase signal. The signal corresponding to the 1 m long drop fibre is considered as the reference for the feeder fibre, and its phase contribution is subtracted from all the others in order to estimate each drop fibre contribution.

To validate the performance of the system, we connected a fibre stretcher to the drop fibres, applying different sinusoidal perturbations. In Fig.2(a) the time traces of the reconstructed phase of feeder fibre and of the drop fibres are shown when a 0.5 V 20 Hz sinusoidal perturbation is applied to the 11 m long drop fibre (called Drop 2). Even if the feeder fibre, which is located in a densely populated area of Athens, is subjected to a strong environmental noise, the sinusoidal perturbation applied to the drop fibre, which is lower in amplitude and in the same band of the environmental noise coming from feeder fibre, is clearly reconstructed. We verified the proof of concept for all drop fibres. As a second example we show in Fig.2(b) the case where a 0.5V 110 Hz perturbation is applied to the 21 m drop fibre (Drop 3). The noise floor has been accurately estimated from the phase noise PSD

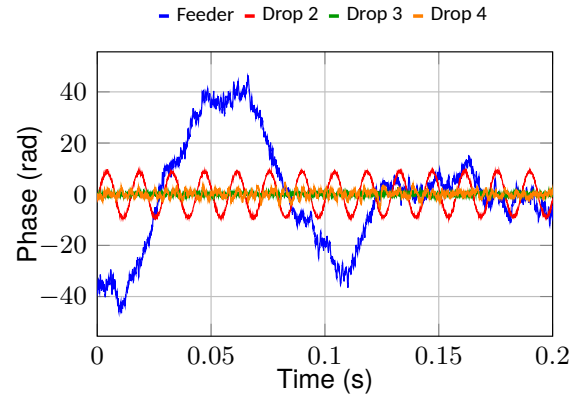


Fig. 3: Drop fibre identification for 1:32 splitter. A sinusoidal perturbation is applied to drop fibre 2. Also for this setup the perturbation is correctly reconstructed.

for each drop fibre signal and varies from 0.1 to 0.2 rad. The minimum detectable elongation for these RMS phase values is below $0.05 \mu\text{m}$, which is more than sufficient to detect anomalies in the infrastructure and generate early warning signals for potential damage to feeder or drop fibres. The adjacent interferometric channels do not show presence of cross-talk, meaning that with the current setup it is possible to achieve excellent isolation of the drop fibres with a few meters of difference between their lengths. We also investigated the capability of the system to operate in a more challenging scenario, by using a 1x32 splitter. In this second case the single pass cumulative losses of feeder fiber and splitter are 26 dB (52 dB bidirectional loss). We used 4 drop fibres having the length of 1 m, 17 m, 300 m, and 2 km, respectively. To overcome the higher losses the pulse length is increased to 50 ns (5 meters resolution). We performed measurements of applied vibrations on the drop fibres. In Fig.3 the reconstructed phase for a 70 Hz sinusoidal perturbation applied to the 17 m drop fibre (Drop 2) is shown. It is evident that the applied perturbations are correctly reconstructed without the presence of cross talk coming from the noisy feeder fibre. The proposed technique can be implemented using low-cost analog and digital electronics and enables near real-time detection of anomalies in feeder and drop fibres, which is essential for early warning and damage prevention in PONs.

Conclusions

We presented a low-complexity cost-effective pulsed interferometry scheme for simultaneous, unambiguous monitoring of PON drop fibres, demonstrated over a 30 km deployed fiber affected by real environmental noise, with a 1:32 split ratio and over 26 dB loss per pass (52 dB total). The technique requires only an FBG at the ONU and a few meters of length difference between drop fibres to accurately identify drop-specific events.

Acknowledgements

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